

Solubility and Diffusion Coefficient of Adenosine 3':5'-Monophosphate*

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Several previously unavailable parameters of adenosine 3':5'-monophosphate have been determined. The molar extinction coefficient at pH 7.0 is 1.38×10^4 , the aqueous solubility at pH 7.0 is 0.0236 M and the diffusion coefficient is 4.44×10^{-6} cm²/s.

In attempting to develop an experimental chemotactic system in which the spatial and temporal dependence of the concentration of adenosine 3':5'-monophosphate (cAMP) were well characterized, we found that certain necessary data, namely, the solubility and diffusion coefficient of the cAMP, were not available. For example, other investigators concerned with the role of cAMP as a diffusible morphogen (1) and as a chemotactic agent for polymorphonucleocytes (2), had been forced to rely on estimates of these parameters. Since interest in this compound is likely to continue to grow because of its ubiquity and importance, we have determined both the aqueous solubility and diffusion coefficient of cAMP experimentally.

MATERIALS AND METHODS

cAMP was the crystalline, free anhydrous acid, 99% pure, obtained from Sigma. Radioactive cAMP was tritiated in carbon 8 and the specific activity was 16 Ci/mmol (Schwarz/Mann).

Absorption spectra and extinction coefficients were determined with a Cary 14 Spectrophotometer. Radioactivity was determined by counting in Bray's solution in a liquid scintillation counter. Under these conditions, the counting efficiency for [³H]cAMP was 55%.

Measurements of the diffusion coefficient were made with the diffusion cell shown in Fig. 1. The cell consists of a Lucite chamber or reservoir with two ports: one to hold the capillary tube containing the test solution and the other to allow the reservoir fluid to be sampled. For each experimental run, a 20- μ l capacity capillary tube (60.5 mm long) was filled completely with a gel composed of 2% Difco agar in distilled water containing the isotopic cAMP. The top of the tube was plugged and the tube was inserted through the port into the 20 ml of distilled water which had been placed in the chamber. The tube opening was located a few millimeters above a magnetic stirrer which kept the reservoir well mixed, and, as a result of dilution in the relatively large reservoir volume, ensured that the concentration of cAMP just outside the tube opening remained negligibly small compared to the concentrations within the tube. Temperature was not controlled, but was constant at $20 \pm 1^\circ$ throughout the run. Triplicate 100- μ l samples were removed at intervals and their radioactivity determined. Concentrations thus obtained were corrected

for cAMP adsorption on the Lucite surfaces of the chamber.

To determine the adsorption correction, the adsorption isotherm of cAMP on Lucite was measured in separate experiments. Various known amounts of [³H]cAMP were added to 20 ml of distilled water in the chamber and the solution was sampled at timed intervals. The amount of [³H]cAMP remaining in solution was determined by counting the samples and each run was continued until there was no further change.

To determine the solubility of cAMP, approximately 10 mg of cAMP was added to 1 ml of phosphate buffer, 10^{-2} M, pH 7.0, and incubated for 2 h at 20°. The solution was then removed and centrifuged at 20° for 2 min at $27,000 \times g$ to remove undissolved solute and any micellar aggregates. The absorbance of the supernatant solution was determined at the absorption maximum ($A_{\max} = 259$ nm) and minimum ($A_{\min} = 227$ nm). Absorbance at these two wavelengths was also determined for a dilute control solution.

RESULTS AND DISCUSSION

The molar extinction coefficient (E_m) for cAMP in 10^{-2} M phosphate buffer, pH 7.0, was determined to be 1.38×10^4 at the absorption maximum, 259 nm. Using this value, the solubility of the cAMP at 20° in this buffer was found to be 0.0236 M.

The adsorption isotherm for cAMP on Lucite is presented in Fig. 2. Approximately one-third of the total activity added to the chamber adsorbed on the surfaces in the concentration range examined so that a substantial correction had to be applied to the volumetric concentration data in calculating the diffusion coefficient. The linear nature of the isotherm is consistent with expectation in this relatively low surface concentration range.

In Fig. 3, increase in [³H]cAMP in the reservoir due to

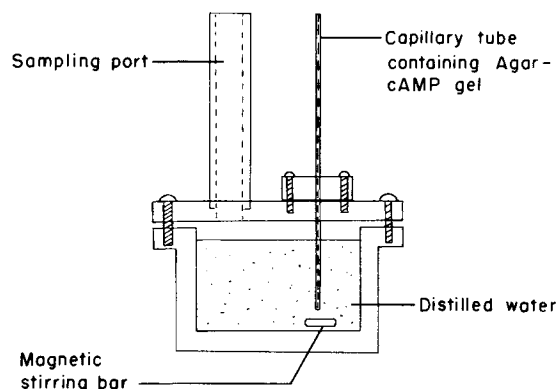


Fig. 1. Diagram of Lucite chamber for diffusion and adsorption measurements.

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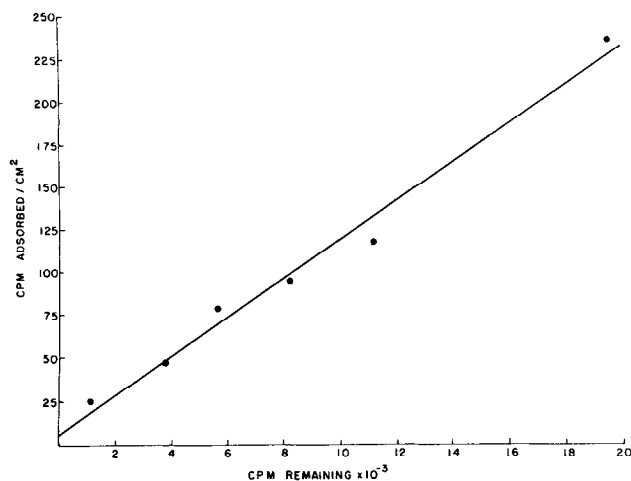


FIG. 2 (left). Adsorption isotherm for [³H]cAMP in diffusion chamber. The data points represent the results of experiments in which 1,965, 3,746, 8,222, 11,137, 15,596, and 27,326 cpm of [³H]cAMP were added to the chamber. Specific activity was 16 Ci/mmol.

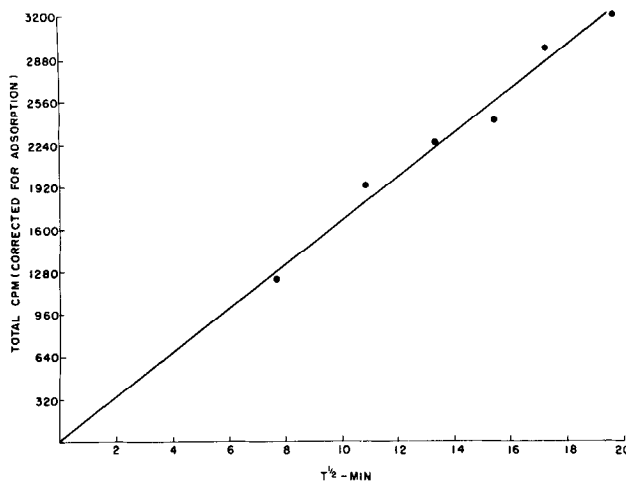


FIG. 3 (right). Diffusion rate of [³H]cAMP (corrected for adsorption). [³H]cAMP (16 Ci/mmol, 0.05 μCi) was added to the capillary tube in 20 μl of 2% Difco agar.

diffusion from the agar-filled capillary tube is plotted as a function of time. The data have been corrected for surface adsorption.

For these experimental conditions, in which the capillary tube is long compared to the "diffusion-penetration depth" (3) and the open end of the capillary tube is maintained at a concentration negligible compared to the initial concentration in the tube, the total amount of cAMP in the reservoir, N , at any time, t , is given by (3):

$$N = 2AC_0\sqrt{\frac{Dt}{\pi}} \quad (1)$$

where A is the area of the capillary tube opening, C_0 is the initial activity of [³H]cAMP in the capillary tube (counts per min per ml), and D is the diffusion coefficient of cAMP in the agar. Thus, the slope of the line in Fig. 3 is $2AC_0\sqrt{D/\pi}$ and

$$D = \frac{\pi}{4(AC_0)^2} \times \text{slope}^2 \quad (2)$$

Based on the data of Fig. 3, the diffusion coefficient is $4.44 \times 10^{-6} \text{ cm}^2/\text{s}$.

The Stokes-Einstein equation (4) gives the following theoretical estimate of the diffusion coefficient of spherical molecules.

$$D = \frac{kT}{6\pi\eta a} \quad (3)$$

where k is the Boltzmann constant, T the temperature, η the solvent viscosity, and a the radius of the molecule. Based on the molecular weight of cAMP, one may estimate its equivalent spherical radius as 5.16 Å, so that Equation 3 predicts a diffusion coefficient of $4.14 \times 10^{-6} \text{ cm}^2/\text{s}$ at 20°, in close agreement with the measured value.

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